# Introduction to Computer Science
# Fall 2026
# Homework 4

**Due: October 25, 2026 at 23:58:59**
**No Copy-and-Paste. No Late Submission.**

**Instructions.** Please write your answers to each problem using LaTeX and submit the compiled PDF file. If you find a problem difficult to understand or solve, you may use external resources, including search engines (such as Google), online references (such as Wikipedia), and AI tools (such as DeepSeek), to study relevant concepts and assist your learning. However, you are responsible for understanding your submitted solution, and **please make sure every word in your submission is written by yourself to avoid plagiarism**. If you use any AI tool while working on the assignment, please include a short disclosure at the end of your submission stating which tool you used and briefly describing how you used it. Undisclosed use of AI tools may be subject to point deductions.

## 1 Local Functions and the Sheaf Property (5 points)

Let $X$ be a finite set and $S = \{0, 1\}$. For each subset $U \subseteq X$, define $\mathcal{F}(U) = \{f : U \to S\}$. Thus, $\mathcal{F}(U)$ contains all functions assigning either 0 or 1 to each element of $U$.

For $V \subseteq U$, the *restriction* $f|_V$ is obtained by keeping only the values of $f$ on $V$.

A family of subsets $U_1, \ldots, U_k$ is a *cover* of $U$ if $U = \bigcup_{i=1}^{k} U_i$.

A family of local functions $f_i \in \mathcal{F}(U_i)$ is called *compatible* if $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for every pair $i, j$.

A *global gluing* is a function $f \in \mathcal{F}(U)$ satisfying $f|_{U_i} = f_i$ for every $i$.

The *sheaf property* requires that every compatible family over every cover has exactly one global gluing.

(a) Let $X = \{a, b, c, d, e\}$, and consider

$$U_1 = \{a, b, c\}, \quad U_2 = \{b, c, d\}, \quad U_3 = \{c, d, e\}.$$

The local functions are

$$f_1 : (a, b, c) \mapsto (1, 0, 1),$$
$$f_2 : (b, c, d) \mapsto (0, 1, 0),$$
$$f_3 : (c, d, e) \mapsto (1, 1, 0).$$

Are these functions compatible? If not, identify every conflicting variable and the corresponding pair of local functions.

(b) Replace $f_3$ with $f_3' : (c, d, e) \mapsto (1, 0, 0)$.

Construct the unique global gluing $f : X \to S$, or explain why none exists.

(c) Consider a different collection:

$$\mathcal{G}(U) = \{f : U \to S \mid f \text{ is constant}\}.$$

A function is *constant* if it assigns the same value to every element of its domain.

For $X = \{a, b\}$, construct a cover $\{U_1, U_2\}$ and compatible local functions $g_i \in \mathcal{G}(U_i)$ that cannot be glued into a global function in $\mathcal{G}(X)$.

Specify the subsets and all function values.

(d) For a cover $\{U_1, \ldots, U_k\}$, define its *overlap graph*: each $U_i$ is a vertex, and distinct vertices $i, j$ are adjacent if $U_i \cap U_j \neq \varnothing$.

Consider two covers of $X = \{a, b, c, d\}$:

$$\begin{aligned} \mathcal{U}_A: &\quad U_1 = \{a, b\},\ U_2 = \{b, c\},\ U_3 = \{c, d\}, \\ \mathcal{U}_B: &\quad V_1 = \{a, b\},\ V_2 = \{c\},\ V_3 = \{d\}. \end{aligned}$$

For each cover, determine:

(i) The number of connected components in its overlap graph.
(ii) The number of compatible families of local constant functions taking values in $S = \{0, 1\}$.

(e) Return to $\mathcal{F}(U) = \{f : U \to S\}$.

Prove that every compatible family of local functions over a cover of $U$ has a unique global gluing.

Your proof should establish both existence and uniqueness.

## 2 Recovering an Object from Its Relations (5 points)

A category has objects and morphisms (arrows). Write $\mathrm{Hom}(X, A)$ for the set of morphisms from $X$ to $A$. If $u : Y \to X$ and $h : X \to A$, their composite $h \circ u : Y \to A$ means applying $u$ first and then $h$. Composition is associative: $(k \circ h) \circ u = k \circ (h \circ u)$ whenever the arrows match. Each object $X$ has an identity $\mathrm{id}_X : X \to X$; for $h : X \to A$,

$$h \circ \mathrm{id}_X = h = \mathrm{id}_A \circ h.$$

Sets and functions form one example of a category. In general, objects need not be sets and morphisms need not be functions. Your general proofs should use composition and identities, without assuming that the objects have elements.

Fix objects $A, B$. For every object $X$, suppose we have an ordinary function

$$\eta_X : \mathrm{Hom}(X, A) \to \mathrm{Hom}(X, B).$$

It takes an entire arrow $h : X \to A$ as input and returns an arrow $\eta_X(h) : X \to B$. The subscript specifies the starting object. Call this family **compatible** if, for every $u : Y \to X$ and $h : X \to A$,

$$\eta_Y(h \circ u) = \eta_X(h) \circ u. \qquad (*)$$

This means that attaching $u$ before translating gives the same result as translating first and then attaching the same $u$. Both results are arrows $Y \to B$. This requirement is called a naturality condition.

(a) Given a fixed $f : A \to B$, define $\eta_X(h) = f \circ h$ for every object $X$ and arrow $h : X \to A$. Check the source and target of $f \circ h$, then verify (*) using associativity. The same definition applies with any starting object: for $k : Y \to A$, we have $\eta_Y(k) = f \circ k$, using the same $f$.

(b) Conversely, suppose a family $\eta$ satisfies (*). We will show that there is a unique $f : A \to B$ with $\eta_X(h) = f \circ h$ for every $X$ and $h : X \to A$. Follow these steps.

(i) Identify a morphism $A \to A$ whose existence is guaranteed by the definition of a category. Use this morphism and a suitable member of the family $\eta$ to construct an arrow $f : A \to B$. State your construction and explain why its source and target are correct.

(ii) Using your construction from (i), prove that $\eta_X(h) = f \circ h$ for every object $X$ and every morphism $h : X \to A$, and that $f$ is the unique morphism $A \to B$ with this property.

(c) An arrow $f : A \to B$ is an **isomorphism** if there is an arrow $g : B \to A$ such that

$$g \circ f = \mathrm{id}_A \qquad \text{and} \qquad f \circ g = \mathrm{id}_B.$$

Then $g$ is an inverse of $f$, and the objects $A, B$ are called isomorphic. Intuitively, going from $A$ to $B$ and back has the same effect as doing nothing; the same holds in the other direction. Both equations are required.

Recall that a function is **injective** if equal outputs imply equal inputs, and **surjective** if every element of the target set has a preimage. It is **bijective** if both properties hold, or equivalently if it has an inverse function.

Let $\eta$ be a compatible family and $f$ its arrow from part (b), so $\eta_X(h) = f \circ h$. We will prove that $f$ is an isomorphism if and only if every $\eta_X$ is bijective.

(i) Suppose $f$ has an inverse $g$. For each $X$, define $\theta_X : \mathrm{Hom}(X, B) \to \mathrm{Hom}(X, A)$ by $\theta_X(k) = g \circ k$. For arbitrary $h : X \to A$ and $k : X \to B$, verify

$$\theta_X(\eta_X(h)) = h \qquad \text{and} \qquad \eta_X(\theta_X(k)) = k.$$

Conclude that $\theta_X$ is the inverse function of $\eta_X$, so $\eta_X$ is bijective.

(ii) Now suppose every $\eta_X$ is bijective. Consider

$$\eta_B : \mathrm{Hom}(B, A) \to \mathrm{Hom}(B, B).$$

Use its surjectivity, with target element $\mathrm{id}_B$, to justify the existence of an arrow $g : B \to A$ satisfying $\eta_B(g) = \mathrm{id}_B$. Use part (b) to deduce $f \circ g = \mathrm{id}_B$.

(iii) The arrows $g \circ f$ and $\mathrm{id}_A$ both belong to $\mathrm{Hom}(A, A)$. Use part (b) and the equation from (ii) to show that their images under $\eta_A$ are equal. Then use the injectivity of $\eta_A$ to deduce $g \circ f = \mathrm{id}_A$. Explain why (ii) and (iii) together show that $f$ is an isomorphism.

## Declaration of AI Use

Please disclose any use of generative AI tools in completing this homework. For each tool used, briefly state what you used it for. You remain responsible for the correctness and originality of your submission. If you did not use any generative AI tools, remove this section.